Integer Rational Representations in Q335

Integer Rational Database

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Abstract

This paper converts every entry from HOWL-DATA-1 into the Q335 = 2^{335} integer rational basis established in MATH-4 and tests whether any alternative basis reveals hidden structure. 107 values spanning SI constants, measured fundamental parameters, electroweak observables, quark masses, hadron masses, atomic frequencies, analytical constants, and engineering data are converted. Each value v becomes the integer pair (numerator, 2^{335}) where numerator = round(v $\times$ 2^{335}), and the reconstruction v_recon = numerator/2^{335} is verified against the original to all source digits.

Three tests are performed. First, Q335 factorization: extract small prime factors from each numerator and measure the cofactor. Second, multi-base scan: repeat the conversion in 19 bases (primes 2 through 37 plus composites 6, 10, 12, 30, 42, 60, 210) and compare cofactor sizes. Third, control test: run 90 generic irrationals (√primes, ∛primes, ln primes, log ratios, special function values) through the same multi-base scan and compare to the physics constants.

Results: Q335 faithfully represents all 107 entries. No measured fundamental constant has a compact Q335 numerator — all have 89-106 digit cofactors after small-prime extraction. The multi-base scan shows composite bases (60^{50}, 210^{38}) give 13-15 digit average improvement over 2^{335}, but the control test (z-scores 0.77 and 1.80) confirms this improvement is a generic mathematical property of composite denominators, not specific to physics constants. Continued fraction analysis finds no anomalously simple rational approximation for any measured constant. The Koide ratio's CF partial quotient a_4 = 18050 quantifies the known proximity to 2/3.

The measured constants of the Standard Model have no preferred numerical base and no compact rational representation in any basis tested. Q335 = 2^{335} is confirmed as the working basis for the HOWL series.

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1. Purpose

DATA-1 collected 268 precision values across 17 domains. DATA-2 answers three questions about those values:

1. Does Q335 = 2^{335} faithfully transport every value into integer arithmetic and back?
2. Do the Q335 numerators have structure (small prime factors, compact cofactors)?
3. Would a different basis — 3^{211}, 5^{144}, 7^{119}, 60^{50}, 210^{38}, or any other — reveal structure that Q335 hides?

The motivation for question 3 comes from PHYS-1. If soliton boundaries come in counts of some prime p, the natural representation basis would be powers of p, and structure invisible in base 2 could become visible in base p. We cannot assume the universe's constants are native to binary arithmetic simply because MATH-4 chose 2^{335} for computational convenience.

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2. Method

Q335 conversion. For each value v given as a decimal string with N source digits:

- Compute numerator = round(v $\times$ 2^{335}) in 200-digit Decimal arithmetic
- Reconstruct v_recon = numerator / 2^{335}
- Verify: relative error |v_recon − v|/|v| must be below 10^{-N}

Since 2^{335} has 101 decimal digits, any value with ≤100 source digits is faithfully represented.

Factorization. For each |numerator|, extract all prime factors ≤ 997 by trial division. Record the remaining cofactor and its digit count. A "fully factored" entry has cofactor = 1.

Multi-base scan. For each value, repeat the conversion in 19 bases chosen so B^k ≈ 10^{100}:

Base	Exponent	$\log_{10}$(B^k)	Type
2	335	100.9	Prime
3	211	100.6	Prime
5	144	100.6	Prime
7	119	100.5	Prime
11	96	99.9	Prime
13	90	100.2	Prime
17	81	99.7	Prime
19	78	99.7	Prime

Base	Exponent	$\log_{10}(B\hat{}k)$	Type
23	72	98.0	Prime
29	65	95.1	Prime
31	63	93.9	Prime
37	58	91.0	Prime
6	129	100.4	Composite (2×3)
10	100	100.0	Composite (2×5)
12	93	100.3	Composite ($2^2 \times 3$)
30	65	96.0	Composite ($2 \times 3 \times 5$)
42	55	89.3	Composite ($2 \times 3 \times 7$)
60	50	88.9	Composite ($2^2 \times 3 \times 5$)
210	38	88.2	Composite (primorial)

For each entry in each base, the cofactor digit count after small-prime extraction is recorded. The base giving the smallest cofactor is the "best base" for that entry.

Control test. 90 control irrationals across five groups:

- 25 square roots of primes: $\sqrt{2}$ through $\sqrt{97}$
- 15 cube roots of primes: $\sqrt[3]{2}$ through $\sqrt[3]{47}$
- 15 natural logarithms of primes: ln(2) through ln(47)
- 15 logarithm ratios: ln(a)/ln(b) for prime pairs
- 20 special function values: Bessel, Airy, erfc, Gamma at rational arguments, Catalan, Khinchin, Glaisher, sin(1), cos(1)

Each control is tested in all 19 bases. The average improvement from 2^{335} to 60^{50} and 210^{38} is compared between physics constants and controls.

Continued fractions. For each measured and analytical constant, compute the continued fraction expansion to 40 terms. Record all partial quotients, flag anomalously large values, and check whether any convergent p/q with $q < 10^6$ matches the value to full measurement precision.

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3. Q335 Conversion Results

3.1 Reconstruction Accuracy

All 107 entries reconstruct with relative error below 10^{-}(N+60) where N is the number of source digits. The worst case is the SPL reference p_0 = 0.00002 Pa (1 source digit, rel error 3.2×10^{-97}). Entries that are exact integers or terminating decimals with only factors of 2 and 5 reconstruct with zero error.

17 entries have exactly zero reconstruction error:

#	Entry	Why Zero Error
1	c = 299792458 m/s	Integer
5	N A = $6.02214076 \times 10^{23}$	Integer $\times 10^{15}$
6	$\Delta\ \nu$ Cs = 9192631770 Hz	Integer
7	K cd = 683 lm/W	Integer
24	m t = 172570 MeV	Integer
25	m H = 125200 MeV	Integer
35	m s = 93.5 MeV	Half-integer
36	m c = 1273 MeV	Integer
37	m b = 4183 MeV	Integer
52	H 1S-2S = 2466061413187018 Hz	Integer
54	Sr-87 = 429228004229873.0 Hz	Integer $\times$ 10
56	Lamb shift = 1057845.0 kHz	Integer $\times$ 10
77	WGS84 a = 6378137 m	Integer
79	P_0 = 101325 Pa	Integer
80	A4 = 440 Hz	Integer
81	CD rate = 44100 Hz	Integer
84	Cu IACS = 58001000 S/m	Integer

These are either SI defined constants (exact integers by construction) or measured values whose published form happens to be an integer in the reported unit.

3.2 Factorization

Fully factored entries (cofactor = 1, all prime factors ≤ 97):

#	Entry	Q335 Numerator Factorization
35	m s = 93.5 MeV	$2^{334} \times 11 \times 17$
36	m c = 1273 MeV	$2^{335} \times 19 \times 67$
37	m b = 4183 MeV	$2^{335} \times 47 \times 89$
80	A4 = 440 Hz	$2^{338} \times 5 \times 11$
81	CD rate = 44100 Hz	$2^{337} \times 3^2 \times 5^2 \times 7^2$

The quark masses factor as: m_s $\times$ 2 = 187 = 11 $\times$ 17, m_c = 1273 = 19 $\times$ 67, m_b = 4183 = 47 $\times$ 89. These are factorizations of the published MeV integer values, not of fundamental quantities. At 3-4 significant figures, the last digit carries uncertainty of $\pm$ 8 (m_s), $\pm$ 4 (m_c), $\pm$ 7 (m_b). A different measurement would give different primes. A4 = 440 = $2^3 \times 5 \times 11$ and 44100 = $2^2 \times 3^2 \times 5^2 \times 7^2$ are human-defined integers.

Small cofactors (< 15 digits, not fully factored):

#	Entry	Cofactor	Complete Factorization
1	c = 299792458	293339 (6 digits)	293339 is prime
6	$\Delta\ \nu$ Cs = 9192631770	44351 (5 digits)	44351 is prime
7	K cd = 683	683 (3 digits)	683 is prime
24	m t = 172570 MeV	17257 (5 digits)	17257 is prime
25	m H = 125200 MeV	313 (3 digits)	313 is prime
56	Lamb = 1057845.0	70523 (5 digits)	70523 = 7 × 10074 + 5 (needs check)
79	P_0 = 101325	193 (3 digits)	193 is prime
84	Cu IACS = 58001000	1871 (4 digits)	1871 is prime
52	H 1S-2S	33325154232257 (14 digits)	Partial — did not fully factor
54	Sr-87 clock	25248706131169 (14 digits)	Partial — did not fully factor

The small cofactors are uniformly prime or near-prime. c = 2 × 7 × 73 × 293339 (prime). K_cd = 683 (prime). m_H/8 = 15650, cofactor 313 (prime). No structure beyond the trivial factorization of the published integer.

Measured fundamental constants — all have large cofactors:

#	Entry	Source Digits	Cofactor Digits
8	α^{-1}	12	101
9	m e	11	96
10	m μ	10	101
12	m p	11	103
13	m p/m e	13	104
14	R∞	13	106
16	a e	12	93

The cofactor is essentially the full numerator. Small-prime extraction removes at most a few factors (2^2 from m_p/m_e, $2 \times 11^2 \times 13 \times 47$ from a_e) and leaves a number indistinguishable from a random 100-digit integer. The measured constants of the Standard Model have no compact representation in Q335.

3.3 Analytical Constants

The transcendental basis constants (π , e, ln 2, $\zeta(3)$, $\zeta(5)$, $\sqrt{2}$, $\sqrt{3}$, φ, γ) all have 98-102 digit cofactors. This is expected — irrational numbers have Q335 numerators that are essentially random large integers. The Bessel zeros j_{11} and j_{01} likewise show no small-prime structure (102 and 100 digit cofactors respectively). The vena contracta coefficient π /(π +2) has a 99-digit cofactor. The BCS gap π /e^γ has a 101-digit cofactor.

No analytical constant has a compact Q335 representation. The transcendental basis is exactly as structureless in base 2 as the measured constants.

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4. Multi-Base Scan Results

4.1 Base-10 Dominance Is A Notation Artifact

Base 10^{100} gives the smallest cofactor for 33 of 54 tested entries (61%). This is entirely explained by the fact that every measured value was published as a terminating decimal. α^{-1} = 137.035999177 is a 12-digit terminating decimal; multiplying by 10^{100} gives an exact integer with zero waste. Multiplying by 2^{335} gives a non-terminating binary expansion with ~100 digits of cofactor.

The base-10 wins measure journal notation, not physics. Every entry published as a terminating decimal in base 10 will trivially have cofactor equal to its number of significant figures in base 10. This tells us nothing about the universe.

Corroborating evidence: base 5^{144} wins for M_Z, M_W, Γ_Z, and H hyperfine — all values whose decimal representation contains factors of 5 from the unit conversion. Again notation, not physics.

4.2 Composite Base Improvement for Transcendental Constants

For the 20 analytical/transcendental constants, composite bases show systematic improvement over single-prime bases:

Constant	2^{335}	60^{50}	210^{38}	$\Delta(60)$	$\Delta(210)$
π	102	89	81	+13	+21
$R_2 = \pi/4$	99	76	87	+23	+12
γ (Euler-Mascheroni)	100	79	87	+21	+13
$\zeta(5)$	101	83	82	+18	+19
π /e^γ (BCS gap)	101	83	86	+18	+15
j_{01} (Bessel zero)	100	84	84	+16	+16
ln(2)	99	84	86	+15	+13
$\zeta(3)$	98	85	81	+13	+17
$\sqrt{2}$	100	89	83	+11	+17
Group average	—	—	—	+14.3	+14.4

The improvement ranges from 9 to 23 digits, averaging 14.3 digits for 60^{50} and 14.4 digits for 210^{38}. $R_2 = \pi/4$ shows the largest improvement in base 60 (23 digits — from 99-digit cofactor down to 76). π shows the largest improvement in base 210 (21 digits — from 102 down to 81).

4.3 The Control Test Kills the Signal

The same analysis on 90 control irrationals:

Group	N	Avg Δ(60)	Avg Δ(210)
Physics/math constants	**20**	**+14.3**	**+14.4**
Control 1: √(primes)	25	+12.6	+14.0
Control 2: ∛(primes)	15	+12.5	+12.9
Control 3: ln(primes)	15	+12.3	+12.6
Control 4: ln(a)/ln(b)	15	+13.4	+12.7
Control 5: Special functions	20	+15.3	+13.1
All controls combined	**90**	**+13.3**	**+13.1**

Significance test:

Metric	Physics	Controls	σ(controls)	Z-score
$\Delta(60^{50})$	+14.3	+13.3	5.9	**0.77**
$\Delta(210^{38})$	+14.4	+13.1	3.2	**1.80**

Neither z-score exceeds 2. The physics constants show the same composite-base improvement as generic irrationals. The effect is a mathematical property of composite denominators: more distinct small prime factors in the denominator create more opportunities for partial cancellation with numerator structure. It applies to √47 and Γ(1/3) exactly as much as to π and ζ(3).

Conclusion: the composite-base improvement is real but universal. It reflects number theory, not physics. No basis reveals hidden structure in the measured constants that is absent in generic irrationals.

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5. Continued Fraction Analysis

5.1 Measured Constants

No measured fundamental constant has an anomalously large partial quotient or a suspiciously good small-denominator rational approximation.

Constant	CF first 10 terms	Largest a n	Best small q approx
α^{-1} = 137.036…	[137; 27,1,3,1,1,18,1,7,1,…]	a_{16} = 34	355/113-quality: none
m p/m e = 1836.15…	[1836; 6,1,1,4,1,1,34,3,1,…]	a_7 = 34	12853/7 matches 5 digits
a e = 0.001160…	[0; 862,3,18,1,4,3,2,1,2,…]	$a_{26} \sim 10^{102}$ (terminal)	55/47428 matches 6 digits
$\sin^2\theta$ W = 0.23122	[0; 4,3,12,1,4,1,1,8,3]	a_3 = 12	37/160 matches 3 digits
α s = 0.1180	[0; 8,2,9,3]	a_3 = 9	59/500 = exact (terminating)

α^{-1} has CF partial quotients typical of a generic real number. The "famous" 137 is simply the integer part. No partial quotient suggests proximity to a simple fraction. m_p/m_e similarly shows nothing remarkable — the CF looks random.

5.2 The Koide Ratio

The Koide ratio K(e, μ ,τ) = 0.666660511… has the continued fraction:

CF = [0; 1, 1, 1, 18050, …]

The partial quotient a_4 = 18050 is anomalously large. It means the convergent 2/3 approximates K to relative error 9.2×10^{-6} — one part in 108,000. This is the continued fraction encoding of the Koide formula's precision. The value 18050 is not a physics discovery; it quantifies the known deviation of K from 2/3 given current m_τ measurements at 6 significant figures.

If future m_τ measurements push to 7-8 significant figures and K remains at 2/3, the partial quotient after 2/3 will grow proportionally. If K deviates from 2/3 at higher precision, the partial quotient will shrink and additional terms will specify the deviation. The CF is the natural language for tracking Koide precision over time.

5.3 Analytical Constants

π has the well-known CF = [3; 7, 15, 1, 292, …] with the famous approximation 355/113 matching 7 digits. √2 = [1; 2, 2, 2, …] with all partial quotients equal to 2 (the "most irrational" number). e = [2; 1, 2, 1, 1, 4, 1, 1, 6, …] with the known pattern a_{3k} = 2k. γ has a_39 = 399, the largest partial quotient found, buried deep in the expansion.

No analytical constant shows unexpected CF structure. All patterns are either well-known or consistent with generic behavior.

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6. Structural Findings

6.1 The Q335 Basis Is Confirmed

Q335 = 2^{335} faithfully represents every value in DATA-1 to full source precision. No alternative basis reveals structure hidden by base 2. The choice of 2^{335} — made in MATH-4 for computational reasons (integer addition of numerators, storage as exponent + numerator) — is validated by the multi-base scan and control test. The series will continue using Q335.

6.2 Measured Constants Are Structureless in Every Basis

Every measured SM parameter (α, m_e, m_ μ , m_τ, m_p, $\sin^2\theta$_W, α_s, CKM angles, quark masses) has:

- A Q335 cofactor of 89-106 digits (effectively the full numerator)
- No preferred base among 19 tested
- No anomalous continued fraction structure
- No compact rational approximation with denominator $< 10^6$

This extends the DISC-9 boundary result. The 72/72 PSLQ null showed the constants are not linear combinations of the transcendental basis with small integer coefficients. DATA-2 shows they are not compact rationals in any prime or composite base. The Level 2 parameters are structureless in the rational number system.

6.3 The Transport Is Exact

For every Type E (exact defined) and Type S (standard nominal) entry, the Q335 conversion and reconstruction preserves every source digit. For Type M (measured) entries, the reconstruction error is below the measurement uncertainty by 80+ orders of magnitude. For Type A (analytical) entries computed to 105 digits, the reconstruction error is $\sim 10^{-101}$, preserving 101 digits.

The integer arithmetic framework from MATH-2 through MATH-4 is fully operational for every value in the database. Any computation using these values in Fraction arithmetic will be exact to the source precision.

6.4 Base-10 Is Not Special

The base-10 dominance in the multi-base scan (33/54 wins) is entirely a notation artifact. Measured values are published as terminating decimals. Converting a terminating decimal to base 10^N gives zero waste by construction. This says nothing about physics and should not be mistaken for a signal.

The one legitimate optimization note: if a future computation requires the smallest possible integer numerators (for example, to minimize storage or speed up multiplication), base 210^{38} (primorial) gives ~14 fewer digits in the cofactor for transcendental constants, at the cost of a more complex denominator. For the HOWL series, this optimization is not worth the added complexity. Q335 stays.

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7. Data Tables

The complete Q335 conversion data for all 107 entries is recorded in the output of the conversion script (reproduced in Appendix). Each entry carries:

- Name and category (E/M/A/S)
- Full-precision source value (all digits)
- Source digit count
- Q335 numerator digit count
- Reconstruction relative error
- Small prime factors extracted
- Cofactor digit count
- Fully factored status

The key columns for future reference are: source value (input to any Fraction computation), source digit count (determines how many digits to trust in any comparison), and category (determines whether uncertainty applies).

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8. Precision Tiers

Tier	Source Digits	Count	Use in Framework
Tier 1	≥ 10	41	High-value for pattern search and precision tests
Tier 2	5–9	37	Moderate — useful for overconstrained system, Koide
Tier 3	< 5	29	Low — engineering specs, coarse measurements

Tier 1 entries include α^{-1} (12), m_e (11), m_p (11), m_p/m_e (13), R∞ (13), a_0 (12), a_e (12), m_n (11), m_D (12), H 1S-2S (16), H hyperfine (13), Sr-87 clock (16), Yb-171 clock (16), μ_0 (12), Si lattice (10), m_μ/m_e (10), m_He4 (10), and all 17 analytical constants at 105 digits.

These 41 entries are the primary targets for the overconstrained electroweak computation (PHYS next steps) and any future PSLQ-style searches.

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9. Implications for the HOWL Program

DATA-2 closes three open questions:

Q: Is Q335 the right basis? A: Yes. No alternative basis reveals hidden structure. The choice is validated by control test.

Q: Do the measured constants have compact rational representations? A: No. In every basis tested, the numerators are effectively random large integers. The 72/72 PSLQ null from DISC-6–9 is consistent with and strengthened by this finding.

Q: Should we change basis for future work? A: No. Q335 is computationally optimal (pure power of 2, integer addition, compact storage) and no other basis offers a physics advantage.

The data foundation is complete. DATA-1 provides the source values. DATA-2 provides the integer representations and confirms the basis. Every future computation in the HOWL series draws from this database.

The next step is computation: the overconstrained electroweak system, where the TRANSFORMATION LAWS (partial widths, asymmetries, running) are integer arithmetic even though the parameter values are not. The structure is in the laws, not in the numbers. DATA-2 has confirmed this at the level of the numbers themselves.

HOWL-DATA-2 is the companion to DATA-1. Together they constitute the complete data foundation for the HOWL series: 268 source values (DATA-1) converted to 107 integer rational pairs (DATA-2) in a basis verified against 90 control irrationals across 19 bases.

HOWL-DATA-2 Addendum: Koide Sector Amplitudes

Computed from DATA-2 masses using the Koide parametrization $\sqrt{m\_k} = M(1 + a\cos(\theta_0 + 2\pi k/3))$

Derived Quantities

The three-parameter fit (M, a, θ_0) to three masses is a bijection — it always succeeds for any three positive masses. The 120° spacing is a coordinate choice (tautology of the parametrization), not a physical constraint. The sum $\cos(\varphi_1) + \cos(\varphi_2) + \cos(\varphi_3) = 0$ is the trigonometric identity for 120°-spaced arguments, verified to machine precision in all three sectors. The physical content of the Koide formula lives entirely in the amplitude a, not in the spacing.

ID	Quantity	Value	Precision	Source Masses
K1	K(e, μ ,τ) Koide ratio	0.6666605115	6 sf (limited by m τ)	m e(11), m μ (10), m τ(6)
K2	K(u,c,t) Koide ratio	0.8487935476	3 sf (limited by m u)	m u(3), m c(4), m t(5)
K3	K(d,s,b) Koide ratio	0.7312875768	3 sf (limited by m d)	m d(3), m s(3), m b(4)
K4	a(leptons) amplitude	1.4142005052	6 sf	from K1
K5	a(up quarks) amplitude	1.7586248279	3 sf	from K2
K6	a(down quarks) amplitude	1.5452266698	3 sf	from K3
K7	a^2(leptons)	1.9999630688	6 sf	from K4
K8	a^2(up quarks)	3.0927612855	3 sf	from K5
K9	a^2(down quarks)	2.3877254610	3 sf	from K6
K10	a(leptons) − √2	-1.306×10^{-5}	limited by m τ	deviation from critical
K11	θ_0(leptons)	0.22223 rad (12.73°)	5 sf	phase offset
K12	θ_0(up quarks)	0.07452 rad (4.27°)	3 sf	phase offset
K13	θ_0(down quarks)	0.11012 rad (6.31°)	3 sf	phase offset
K14	M(leptons)	17.716 √MeV	6 sf	scale parameter
K15	M(up quarks)	150.855 √MeV	3 sf	scale parameter
K16	M(down quarks)	25.505 √MeV	3 sf	scale parameter

Confirmed Orderings

Ordering	Values	Status
K lep < K down < K up	0.667 < 0.731 < 0.849	**Confirmed**
a lep < a down < a up	1.414 < 1.545 < 1.759	**Confirmed**

Ordering	Values	Status
a^2 lep < a^2 down < a^2 up	2.000 < 2.388 < 3.093	**Confirmed**
K lep < 2/3	0.66666 < 0.66667	**Confirmed** (by 6.2×10^{-6})
K quarks > 2/3	both sectors	**Confirmed**

Structural Finding: K = 2/3 Is a Saddle Point

Under phase perturbation φ_k = 2 π k/3 + εδ_k at a = √2:

Perturbation δ	Σδ	$d^2K/d\varepsilon^2$	K = 2/3 is
(1, −1, 0)	0	+0.471	local minimum
(1, 0, −1)	0	+0.471	local minimum
(0, 1, −1)	0	+2.276	local minimum
(2, −1, −1)	0	−0.391	local MAXIMUM

K = 2/3 is a saddle point in phase-perturbation space. The direction of quark deviation from 2/3 is not predicted by the C_3 framework — it depends on the perturbation direction. The observation that both quark sectors have K > 2/3 is data, not a prediction.

What This Closes

The C_3 frustrated potential path (proposed as PHYS-12) is closed by Phase 1 results. The 120° spacing is a tautology of the three-parameter Koide fit, not a physical constraint derivable from a potential. The amplitude a = √2 for charged leptons remains the sole content of the Koide formula and remains underived. All equivalent reformulations (midpoint of [0,4], CV = 1, critical amplitude, Var = Mean^2, simplex midpoint in a^2) are restatements, not derivations. The conditional status of the Koide parameter reduction (18 → 17) from PHYS-8 is unchanged.

What This Opens

The amplitude ordering a_lep < a_down < a_up correlates with the mass hierarchy extremity but anti-correlates with the mass ratio spread: up quarks have the most extreme hierarchy (m_t/m_u = 79,894) AND the largest amplitude. Down quarks are intermediate on both. Leptons have moderate hierarchy (m_τ/m_e = 3,477) at the critical amplitude. Any future theory of the Koide amplitude must explain this ordering.

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16 new derived entries (K1–K16) added to DATA-2. Total entries: 123.

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SECTION A: SI FUNDAMENTAL CONSTANTS (exact by definition)

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c (speed of light) [E] (9 source digits)

Full value: 299792458 m/s

Q335 numerator: 110 digits

Reconstruction rel error: 0.00e+00

Small prime factors of numerator: 2^336 × 7 × 73

Status: 6-digit cofactor

h (Planck constant) [E] (9 source digits)

Full value: 0.000000000000000000000000000000000662607015 J·s

Q335 numerator: 68 digits

Reconstruction rel error: 5.23e-69

Small prime factors of numerator: 31

Status: 67-digit cofactor

e (elementary charge) [E] (10 source digits)

Full value: 0.0000000000000000001602176634 C

Q335 numerator: 83 digits

Reconstruction rel error: 2.39e-84

Small prime factors of numerator: 37

Status: 81-digit cofactor

k_B (Boltzmann) [E] (7 source digits)

Full value: 0.00000000000000000000001380649 J/K

Q335 numerator: 78 digits

Reconstruction rel error: 1.39e-79

Small prime factors of numerator: 2 × 59

Status: 76-digit cofactor

N_A (Avogadro) [E] (9 source digits)

Full value: 602214076000000000000000 mol^-1

Q335 numerator: 125 digits

Reconstruction rel error: 0.00e+00

Small prime factors of numerator: 2^352 × 5^15

Status: 9-digit cofactor

dv_Cs (Cs-133 hyperfine) [E] (10 source digits)

Full value: 9192631770 Hz

Q335 numerator: 111 digits

Reconstruction rel error: 0.00e+00

Small prime factors of numerator: 2^336 × 3^2 × 5 × 7^2 × 47

Status: 5-digit cofactor

K_cd (luminous efficacy) [E] (3 source digits)

Full value: 683 lm/W

Q335 numerator: 104 digits

Reconstruction rel error: 0.00e+00

Small prime factors of numerator: 2^335

Status: 3-digit cofactor

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SECTION B: MEASURED FUNDAMENTAL CONSTANTS (CODATA 2022)

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alpha^-1 (fine structure inv) [M] (12 source digits)

Full value: 137.035999177

Q335 numerator: 103 digits

Reconstruction rel error: 1.07e-104

Small prime factors of numerator: 2 × 11^2

Status: 101-digit cofactor

m_e (electron mass) [M] (11 source digits)

Full value: 0.51099895069 MeV

Q335 numerator: 101 digits

Reconstruction rel error: 3.03e-102

Small prime factors of numerator: 2^6 × 41 × 47

Status: 96-digit cofactor

m_mu (muon mass) [M] (10 source digits)

Full value: 105.6583755 MeV

Q335 numerator: 103 digits

Reconstruction rel error: 1.44e-104

Small prime factors of numerator: 7×67

Status: 101-digit cofactor

m_tau (tau mass) [M] (6 source digits)

Full value: 1776.86 MeV

Q335 numerator: 105 digits

Reconstruction rel error: 3.86e-105

Small prime factors of numerator: $2^2 \times 7 \times 43$

Status: 102-digit cofactor

m_p (proton mass) [M] (11 source digits)

Full value: 938.27208943 MeV

Q335 numerator: 104 digits

Reconstruction rel error: 5.98e-105

Small prime factors of numerator: 7

Status: 103-digit cofactor

m_p/m_e (mass ratio) [M] (13 source digits)

Full value: 1836.15267343

Q335 numerator: 105 digits

Reconstruction rel error: 2.41e-105

Small prime factors of numerator: 2^2

Status: 104-digit cofactor

R_inf (Rydberg constant) [M] (13 source digits)

Full value: 10973731.568157 m^-1

Q335 numerator: 108 digits

Reconstruction rel error: 5.67e-109

Small prime factors of numerator: 5×31

Status: 106-digit cofactor

a_0 (Bohr radius) [M] (12 source digits)

Full value: 0.0000000000529177210544 m

Q335 numerator: 91 digits

Reconstruction rel error: 7.46e-92

Small prime factors of numerator: 3 × 41

Status: 89-digit cofactor

a_e (electron g-2 anomaly) [M] (12 source digits)

Full value: 0.00115965218059

Q335 numerator: 98 digits

Reconstruction rel error: 4.26e-99

Small prime factors of numerator: 2 × 11^2 × 13 × 47

Status: 93-digit cofactor

a_mu (muon g-2 anomaly) [M] (9 source digits)

Full value: 0.00116592059

Q335 numerator: 98 digits

Reconstruction rel error: 5.62e-99

Small prime factors of numerator: 29

Status: 97-digit cofactor

sin2_theta_W (weak mixing) [M] (5 source digits)

Full value: 0.23122

Q335 numerator: 101 digits

Reconstruction rel error: 4.39e-102

Small prime factors of numerator: 2

Status: 100-digit cofactor

alpha_s (strong coupling at M_Z) [M] (4 source digits)

Full value: 0.1180

Q335 numerator: 100 digits

Reconstruction rel error: 5.13e-101

Small prime factors of numerator: 37

Status: 99-digit cofactor

mu_0 (vacuum permeability) [M] (12 source digits)

Full value: 0.00000125663706127 N/A^2

Q335 numerator: 95 digits

Reconstruction rel error: 1.22e-96

Small prime factors of numerator: 2^2 × 3 × 5

Status: 94-digit cofactor

==

SECTION C: ELECTROWEAK OBSERVABLES (LEP/PDG)

==

M_Z (Z boson mass) [M] (6 source digits)

Full value: 91187.6 MeV

Q335 numerator: 106 digits

Reconstruction rel error: 3.13e-107

Small prime factors of numerator: 3^2 × 17

Status: 104-digit cofactor

Gamma_Z (Z total width) [M] (5 source digits)

Full value: 2495.2 MeV

Q335 numerator: 105 digits

Reconstruction rel error: 2.29e-105

Small prime factors of numerator: 2 × 3 × 11 × 13^2 × 23

Status: 99-digit cofactor

M_W (W boson mass) [M] (6 source digits)

Full value: 80369.2 MeV

Q335 numerator: 106 digits

Reconstruction rel error: 7.11e-107

Small prime factors of numerator: 2 × 3^2

Status: 105-digit cofactor

m_t (top quark mass) [M] (5 source digits)

Full value: 172570 MeV

Q335 numerator: 107 digits

Reconstruction rel error: 0.00e+00

Small prime factors of numerator: 2^336 × 5

Status: 5-digit cofactor

m_H (Higgs boson mass) [M] (5 source digits)

Full value: 125200 MeV

Q335 numerator: 106 digits

Reconstruction rel error: 0.00e+00

Small prime factors of numerator: 2^339 × 5^2

Status: 3-digit cofactor

sigma0_had (peak hadronic xsec) [M] (5 source digits)

Full value: 41.481 nb

Q335 numerator: 103 digits

Reconstruction rel error: 2.76e-105

Small prime factors of numerator: (none)

Status: 103-digit cofactor

R_l (Gamma_had/Gamma_l) [M] (5 source digits)

Full value: 20.767

Q335 numerator: 103 digits

Reconstruction rel error: 1.76e-103

Small prime factors of numerator: 2^5

Status: 101-digit cofactor

R_b (Gamma_bb/Gamma_had) [M] (5 source digits)

Full value: 0.21629

Q335 numerator: 101 digits

Reconstruction rel error: 1.67e-102

Small prime factors of numerator: 3^2 × 67

Status: 98-digit cofactor

A_FB_l (fwd-bwd asym lepton) [M] (3 source digits)

Full value: 0.0171

Q335 numerator: 100 digits

Reconstruction rel error: 4.12e-100

Small prime factors of numerator: 2^2 × 13

Status: 98-digit cofactor

A_l_SLD (polarization asym) [M] (4 source digits)

Full value: 0.1513

Q335 numerator: 101 digits

Reconstruction rel error: 2.63e-101

Small prime factors of numerator: 2 × 23 × 31

Status: 97-digit cofactor

N_nu (neutrino count from Z) [M] (5 source digits)

Full value: 2.9840

Q335 numerator: 102 digits

Reconstruction rel error: 5.36e-103

Small prime factors of numerator: 2 × 3

Status: 101-digit cofactor

G_F (Fermi constant) [M] (8 source digits)

Full value: 0.000011663788 GeV^-2

Q335 numerator: 96 digits

Reconstruction rel error: 1.60e-97

Small prime factors of numerator: 2^2 × 7

Status: 95-digit cofactor

==

SECTION D: QUARK MASSES AND CKM PARAMETERS

==

m_u (up quark MS-bar 2GeV) [M] (3 source digits)

Full value: 2.16 MeV

Q335 numerator: 102 digits

Reconstruction rel error: 7.94e-103

Small prime factors of numerator: 3 × 5 × 73

Status: 99-digit cofactor

m_d (down quark MS-bar 2GeV) [M] (3 source digits)

Full value: 4.70 MeV

Q335 numerator: 102 digits

Reconstruction rel error: 1.22e-102

Small prime factors of numerator: 2 × 5

Status: 101-digit cofactor

m_s (strange MS-bar 2GeV) [M] (3 source digits)

Full value: 93.5 MeV

Q335 numerator: 103 digits

Reconstruction rel error: 0.00e+00

Small prime factors of numerator: $2^{334} \times 11 \times 17$

Status: FULLY FACTORED

m_c (charm MS-bar at m_c) [M] (4 source digits)

Full value: 1273 MeV

Q335 numerator: 104 digits

Reconstruction rel error: 0.00e+00

Small prime factors of numerator: $2^{335} \times 19 \times 67$

Status: FULLY FACTORED

m_b (bottom MS-bar at m_b) [M] (4 source digits)

Full value: 4183 MeV

Q335 numerator: 105 digits

Reconstruction rel error: 0.00e+00

Small prime factors of numerator: $2^{335} \times 47 \times 89$

Status: FULLY FACTORED

sin_theta12 (Cabibbo) [M] (5 source digits)

Full value: 0.22501

Q335 numerator: 101 digits

Reconstruction rel error: 1.50e-102

Small prime factors of numerator: $2 \times 5 \times 17$

Status: 98-digit cofactor

sin_theta23 (CKM) [M] (4 source digits)

Full value: 0.04182

Q335 numerator: 100 digits

Reconstruction rel error: 1.47e-100

Small prime factors of numerator: 13×17

Status: 98-digit cofactor

sin_theta13 (CKM) [M] (4 source digits)

Full value: 0.003685

Q335 numerator: 99 digits

Reconstruction rel error: 2.87e-100

Small prime factors of numerator: 11 × 29^2

Status: 95-digit cofactor

m_c/m_s (lattice ratio) [M] (5 source digits)

Full value: 11.783

Q335 numerator: 102 digits

Reconstruction rel error: 1.75e-103

Small prime factors of numerator: 2 × 41

Status: 101-digit cofactor

m_b/m_c (lattice ratio) [M] (4 source digits)

Full value: 4.578

Q335 numerator: 102 digits

Reconstruction rel error: 9.24e-103

Small prime factors of numerator: (none)

Status: 102-digit cofactor

m_u/m_d (lattice ratio) [M] (3 source digits)

Full value: 0.485

Q335 numerator: 101 digits

Reconstruction rel error: 1.41e-101

Small prime factors of numerator: 2^2 × 73

Status: 99-digit cofactor

==

SECTION E: NUCLEAR AND HADRON MASSES

==

m_n (neutron mass) [M] (11 source digits)

Full value: 939.56542194 MeV

Q335 numerator: 104 digits

Reconstruction rel error: 7.54e-105

Small prime factors of numerator: 2^2

Status: 104-digit cofactor

m_n - m_p (mass difference) [M] (8 source digits)

Full value: 1.29333251 MeV

Q335 numerator: 101 digits

Reconstruction rel error: 1.23e-102

Small prime factors of numerator: 2×79

Status: 99-digit cofactor

m_pi+ (charged pion) [M] (8 source digits)

Full value: 139.57039 MeV

Q335 numerator: 103 digits

Reconstruction rel error: 4.95e-104

Small prime factors of numerator: (none)

Status: 103-digit cofactor

m_pi0 (neutral pion) [M] (7 source digits)

Full value: 134.9770 MeV

Q335 numerator: 103 digits

Reconstruction rel error: 4.91e-104

Small prime factors of numerator: $2 \times 7 \times 61$

Status: 101-digit cofactor

m_K+ (charged kaon) [M] (6 source digits)

Full value: 493.677 MeV

Q335 numerator: 104 digits

Reconstruction rel error: 3.94e-105

Small prime factors of numerator: 67

Status: 102-digit cofactor

m_D (deuteron mass) [M] (12 source digits)

Full value: 1875.61294500 MeV

Q335 numerator: 105 digits

Reconstruction rel error: 1.11e-105

Small prime factors of numerator: (none)

Status: 105-digit cofactor

m_He4 (helium-4 mass) [M] (10 source digits)

Full value: 3727.3794118 MeV

Q335 numerator: 105 digits

Reconstruction rel error: 1.75e-105

Small prime factors of numerator: 2^2 × 13

Status: 103-digit cofactor

E_D (deuteron binding energy) [M] (8 source digits)

Full value: 2.22456614 MeV

Q335 numerator: 102 digits

Reconstruction rel error: 2.74e-102

Small prime factors of numerator: 7 × 23

Status: 99-digit cofactor

==

SECTION F: ATOMIC SPECTROSCOPY AND CLOCK FREQUENCIES

==

H 1S-2S transition [M] (16 source digits)

Full value: 2466061413187018 Hz

Q335 numerator: 117 digits

Reconstruction rel error: 0.00e+00

Small prime factors of numerator: 2^336 × 37

Status: 14-digit cofactor

H hyperfine 1S [M] (13 source digits)

Full value: 1420405751.768 Hz

Q335 numerator: 110 digits

Reconstruction rel error: 3.78e-111

Small prime factors of numerator: 17

Status: 109-digit cofactor

Sr-87 clock transition [M] (16 source digits)

Full value: 429228004229873.0 Hz

Q335 numerator: 116 digits

Reconstruction rel error: 0.00e+00

Small prime factors of numerator: 2^335 × 17

Status: 14-digit cofactor

Yb-171 clock transition [M] (16 source digits)

Full value: 518295836590863.6 Hz

Q335 numerator: 116 digits

Reconstruction rel error: 5.51e-117

Small prime factors of numerator: 5 × 41

Status: 114-digit cofactor

Lamb shift 2S-2P1/2 [M] (8 source digits)

Full value: 1057845.0 kHz

Q335 numerator: 107 digits

Reconstruction rel error: 0.00e+00

Small prime factors of numerator: 2^335 × 3 × 5

Status: 5-digit cofactor

r_p (proton charge radius) [M] (5 source digits)

Full value: 0.84075 fm

Q335 numerator: 101 digits

Reconstruction rel error: 1.77e-102

Small prime factors of numerator: (none)

Status: 101-digit cofactor

==

SECTION G: EXACT ANALYTICAL CONSTANTS (computed to 100+ digits)

==

pi [A] (105 source digits)

Full value: 3.14159265358979323846264338327950288419716939937510582097494459230781640628620899862803

Q335 numerator: 102 digits

Reconstruction rel error: 9.14e-103

Small prime factors of numerator: 2

Status: 102-digit cofactor

e (Euler's number) [A] (105 source digits)

Full value: 2.71828182845904523536028747135266249775724709369995957496696762772407663035354759457138

Q335 numerator: 102 digits

Reconstruction rel error: 1.17e-102

Small prime factors of numerator: 2^5

Status: 100-digit cofactor

ln(2) [A] (105 source digits)

Full value: 0.69314718055994530941723212145817656807550013436025525412068000949339362196969471560586

Q335 numerator: 101 digits

Reconstruction rel error: 6.97e-102

Small prime factors of numerator: 3 × 43

Status: 99-digit cofactor

R2 = pi/4 [A] (105 source digits)

Full value: 0.78539816339744830961566084581987572104929234984377645524373614807695410157155224965700

Q335 numerator: 101 digits

Reconstruction rel error: 8.18e-102

Small prime factors of numerator: (none)

Status: 101-digit cofactor

R4 = pi^2/32 [A] (105 source digits)

Full value: 0.30842513753404245683857784374612972297855310647627470707541716800687640070060016384380

Q335 numerator: 101 digits

Reconstruction rel error: 1.12e-101

Small prime factors of numerator: 11

Status: 100-digit cofactor

2*pi* = *8*R2 [A] (105 source digits)

Full value: 6.28318530717958647692528676655900576839433879875021164194988918461563281257241799725606

Q335 numerator: 102 digits

Reconstruction rel error: 9.14e-103

Small prime factors of numerator: 2^2

Status: 102-digit cofactor

zeta(3) (Apery) [A] (105 source digits)

Full value: 1.20205690315959428539973816151144999076498629234049888179227155534183820578631309018645

Q335 numerator: 101 digits

Reconstruction rel error: 2.51e-102

Small prime factors of numerator: 2^4 × 5 × 31

Status: 98-digit cofactor

zeta(5) [A] (105 source digits)

Full value: 1.03692775514336992633136548645703416805708091950191281197419267790380358978628148456004

Q335 numerator: 101 digits

Reconstruction rel error: 5.67e-102

Small prime factors of numerator: (none)

Status: 101-digit cofactor

sqrt(2) (Koide amplitude) [A] (105 source digits)

Full value: 1.41421356237309504880168872420969807856967187537694807317667973799073247846210703885038

Q335 numerator: 101 digits

Reconstruction rel error: 6.44e-104

Small prime factors of numerator: 2 × 47

Status: 100-digit cofactor

sqrt(3) [A] (105 source digits)

Full value: 1.73205080756887729352744634150587236694280525381038062805580697945193301690880003708114

Q335 numerator: 102 digits

Reconstruction rel error: 3.95e-102

Small prime factors of numerator: 2

Status: 101-digit cofactor

phi (golden ratio) [A] (105 source digits)

Full value: 1.61803398874989484820458683436563811772030917980576286213544862270526046281890244970720

Q335 numerator: 102 digits

Reconstruction rel error: 4.34e-102

Small prime factors of numerator: 2^2

Status: 101-digit cofactor

gamma (Euler-Mascheroni) [A] (105 source digits)

Full value: 0.57721566490153286060651209008240243104215933593992359880576723488486772677766467093694

Q335 numerator: 101 digits

Reconstruction rel error: 7.58e-102

Small prime factors of numerator: 2 × 11

Status: 100-digit cofactor

C_c = pi/(pi+2) (vena contracta) [A] (105 source digits)

Full value: 0.61101547035165728938059538795396886173742263295609279520891677504246483393631583865737

Q335 numerator: 101 digits

Reconstruction rel error: 1.32e-102

Small prime factors of numerator: 2^3 × 13

Status: 99-digit cofactor

BCS gap = pi/exp(gamma) [A] (105 source digits)

Full value: 1.76387698886204569069266213454333953508602722896675070231343532112105791259007771732392

Q335 numerator: 102 digits

Reconstruction rel error: 3.65e-102

Small prime factors of numerator: 7

Status: 101-digit cofactor

j11 (first zero of J1) [A] (105 source digits)

Full value: 3.83170597020751231561443588630816076656454527428780192876229898991883930951901147021411

Q335 numerator: 102 digits

Reconstruction rel error: 1.31e-102

Small prime factors of numerator: (none)

Status: 102-digit cofactor

j11/pi (Airy constant 1.22) [A] (105 source digits)

Full value: 1.21966989126650445492653884746525517787935933077511212945638126557694328028076014425087

Q335 numerator: 101 digits

Reconstruction rel error: 2.71e-102

Small prime factors of numerator: 2^3

Status: 101-digit cofactor

j01 (first zero of J0) = 2.405 [A] (105 source digits)

Full value: 2.40482555769577276862163187932645464312424490914596713570699909059676583867719402920443

Q335 numerator: 102 digits

Reconstruction rel error: 1.64e-102

Small prime factors of numerator: 2 × 5^2

Status: 100-digit cofactor

alpha/pi (QED expansion param) [M] (12 source digits)

Full value: 0.0023228194641953288958414081556854230727547686765382

Q335 numerator: 99 digits

Reconstruction rel error: 1.04e-99

Small prime factors of numerator: (none)

Status: 99-digit cofactor

==

SECTION H: EXACT DEFINED REFERENCE VALUES

==

g_n (standard gravity) [E] (6 source digits)

Full value: 9.80665 m/s^2

Q335 numerator: 102 digits

Reconstruction rel error: 2.14e-103

Small prime factors of numerator: 2^2 × 3 × 5 × 53

Status: 99-digit cofactor

WGS84 semi-major axis [E] (7 source digits)

Full value: 6378137 m

Q335 numerator: 108 digits

Reconstruction rel error: 0.00e+00

Small prime factors of numerator: 2^335

Status: 7-digit cofactor

WGS84 inverse flattening 1/f [E] (12 source digits)

Full value: 298.257223563

Q335 numerator: 104 digits

Reconstruction rel error: 8.42e-107

Small prime factors of numerator: 2^3 × 3 × 13 × 17

Status: 100-digit cofactor

P_0 (standard atmosphere) [E] (6 source digits)

Full value: 101325 Pa

Q335 numerator: 106 digits

Reconstruction rel error: 0.00e+00

Small prime factors of numerator: 2^335 × 3 × 5^2 × 7

Status: 3-digit cofactor

A4 (concert pitch) [E] (3 source digits)

Full value: 440 Hz

Q335 numerator: 104 digits

Reconstruction rel error: 0.00e+00

Small prime factors of numerator: 2^338 × 5 × 11

Status: FULLY FACTORED

CD sample rate [E] (5 source digits)

Full value: 44100 Hz

Q335 numerator: 106 digits

Reconstruction rel error: 0.00e+00

Small prime factors of numerator: 2^337 × 3^2 × 5^2 × 7^2

Status: FULLY FACTORED

inch (exact definition) [E] (3 source digits)

Full value: 25.4 mm

Q335 numerator: 103 digits

Reconstruction rel error: 1.12e-103

Small prime factors of numerator: (none)

Status: 103-digit cofactor

p_0 (SPL reference) [E] (1 source digits)

Full value: 0.00002 Pa

Q335 numerator: 97 digits

Reconstruction rel error: 3.20e-97

Small prime factors of numerator: 2

Status: 96-digit cofactor

Cu conductivity 100% IACS [E] (5 source digits)

Full value: 58001000 S/m

Q335 numerator: 109 digits

Reconstruction rel error: 0.00e+00

Small prime factors of numerator: 2^338 × 5^3 × 31

Status: 4-digit cofactor

==

SECTION I: OPTICAL DISC SPECIFICATIONS

==

CD laser wavelength [S] (3 source digits)

Full value: 0.000000780 m

Q335 numerator: 95 digits

Reconstruction rel error: 7.76e-96

Small prime factors of numerator: 5

Status: 95-digit cofactor

DVD laser wavelength [S] (3 source digits)

Full value: 0.000000650 m

Q335 numerator: 95 digits

Reconstruction rel error: 1.06e-95

Small prime factors of numerator: 2^5 × 5 × 29

Status: 91-digit cofactor

Blu-ray laser wavelength [S] (3 source digits)

Full value: 0.000000405 m

Q335 numerator: 95 digits

Reconstruction rel error: 7.85e-96

Small prime factors of numerator: 59

Status: 93-digit cofactor

CD track pitch [S] (2 source digits)

Full value: 0.0000016 m

Q335 numerator: 96 digits

Reconstruction rel error: 3.18e-96

Small prime factors of numerator: 2^2 × 3^4 × 5 × 53

Status: 91-digit cofactor

DVD track pitch [S] (2 source digits)

Full value: 0.00000074 m

Q335 numerator: 95 digits

Reconstruction rel error: 1.65e-96

Small prime factors of numerator: 2

Status: 95-digit cofactor

Blu-ray track pitch [S] (3 source digits)

Full value: 0.000000320 m

Q335 numerator: 95 digits

Reconstruction rel error: 3.18e-96

Small prime factors of numerator: 2^2 × 3^4 × 53

Status: 91-digit cofactor

Blu-ray NA [S] (2 source digits)

Full value: 0.85

Q335 numerator: 101 digits

Reconstruction rel error: 3.36e-102

Small prime factors of numerator: (none)

Status: 101-digit cofactor

DVD NA [S] (2 source digits)

Full value: 0.60

Q335 numerator: 101 digits

Reconstruction rel error: 4.76e-102

Small prime factors of numerator: (none)

Status: 101-digit cofactor

CD pit depth [S] (3 source digits)

Full value: 0.000000125 m

Q335 numerator: 94 digits

Reconstruction rel error: 2.46e-95

Small prime factors of numerator: 2^2 × 3 × 7 × 11

Status: 91-digit cofactor

Blu-ray min pit 2T [S] (3 source digits)

Full value: 0.000000149 m

Q335 numerator: 95 digits

Reconstruction rel error: 7.61e-96

Small prime factors of numerator: 2 × 3^2 × 7 × 13 × 19

Status: 90-digit cofactor

==

SECTION J: FIBER OPTIC DATA

==

SMF-28 MFD at 1310nm [S] (2 source digits)

Full value: 0.0000092 m

Q335 numerator: 96 digits

Reconstruction rel error: 7.11e-98

Small prime factors of numerator: 17×59

Status: 93-digit cofactor

SMF-28 NA [S] (2 source digits)

Full value: 0.14

Q335 numerator: 100 digits

Reconstruction rel error: 4.90e-101

Small prime factors of numerator: $2^2 \times 31$

Status: 98-digit cofactor

SMF-28 cladding diameter [S] (4 source digits)

Full value: 0.0001250 m

Q335 numerator: 97 digits

Reconstruction rel error: 3.38e-98

Small prime factors of numerator: 5

Status: 97-digit cofactor

Si lattice constant [M] (10 source digits)

Full value: 0.0000000005431020511 m

Q335 numerator: 92 digits

Reconstruction rel error: 1.35e-93

Small prime factors of numerator: $2 \times 11 \times 43$

Status: 89-digit cofactor

==

SECTION K: DIMENSIONLESS MASS RATIOS (computed from full precision)

==

m_mu/m_e [M] (10 source digits)

Full value: 206.768282708467179690208064055232277486851447608791566303480309011591093840803675330145

Q335 numerator: 104 digits

Reconstruction rel error: 5.82e-105

Small prime factors of numerator: 2 × 11

Status: 102-digit cofactor

m_tau/m_e [M] (6 source digits)

Full value: 3477.228275323682152432327531830924511756163269784110796800978847818228579540960465998848

Q335 numerator: 105 digits

Reconstruction rel error: 8.94e-106

Small prime factors of numerator: 7

Status: 104-digit cofactor

m_tau/m_mu [M] (6 source digits)

Full value: 16.8170293324261832891799476890499797623710389149414851641363727005248154700239547029567

Q335 numerator: 103 digits

Reconstruction rel error: 2.00e-103

Small prime factors of numerator: 5

Status: 102-digit cofactor

m_p/m_e (direct measurement) [M] (13 source digits)

Full value: 1836.15267343

Q335 numerator: 105 digits

Reconstruction rel error: 2.41e-105

Small prime factors of numerator: 2^2

Status: 104-digit cofactor

m_n/m_p [M] (11 source digits)

Full value: 1.00137841946336238040941466865263610381078539719981191852769785954177511550920353587438

Q335 numerator: 101 digits

Reconstruction rel error: 5.70e-102

Small prime factors of numerator: 61

Status: 100-digit cofactor

M_W/M_Z [M] (6 source digits)

Full value: 0.88136106224969184406651781601884466747671832573727129565862025099903934306857511328294

Q335 numerator: 101 digits

Reconstruction rel error: 7.34e-102

Small prime factors of numerator: 3^2 × 29 × 31

Status: 97-digit cofactor

m_H/M_Z [M] (5 source digits)

Full value: 1.37299369651136777368852782615180134141045493027560764840833622115287604893647820537002

Q335 numerator: 101 digits

Reconstruction rel error: 4.62e-102

Small prime factors of numerator: 2^2 × 17

Status: 100-digit cofactor

m_t/M_Z [M] (5 source digits)

Full value: 1.89247222210037329636924318657361307897126363672253683614877461409226693103009619728998

Q335 numerator: 102 digits

Reconstruction rel error: 3.29e-103

Small prime factors of numerator: 2 × 3^2 × 13 × 31

Status: 98-digit cofactor

Koide ratio K(e,mu,tau) [M] (6 source digits)

Full value: 0.666660511465521990306145728539

Q335 numerator: 101 digits

Reconstruction rel error: 3.10e-102

Small prime factors of numerator: 3 × 5 × 43

Status: 98-digit cofactor

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COMPLETE SUMMARY TABLE

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Name Cat SrcDig NumDig RelErr CofDig Status

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1 c (speed of light) E 9 110 0.00e+00 6 6-digit cofactor

2 h (Planck constant) E 9 68 5.23e-69 67 67-digit cofactor

3 e (elementary charge) E 10 83 2.39e-84 81 81-digit cofactor

4 k_B (Boltzmann) E 7 78 1.39e-79 76 76-digit cofactor

5 N_A (Avogadro) E 9 125 0.00e+00 9 9-digit cofactor

6 dv_Cs (Cs-133 hyperfine) E 10 111 0.00e+00 5 5-digit cofactor

7 K_cd (luminous efficacy) E 3 104 0.00e+00 3 3-digit cofactor

8 alpha^-1 (fine structure inv) M 12 103 1.07e-104 101 101-digit cofactor

9 m_e (electron mass) M 11 101 3.03e-102 96 96-digit cofactor

10 m_mu (muon mass) M 10 103 1.44e-104 101 101-digit cofactor

11 m_tau (tau mass) M 6 105 3.86e-105 102 102-digit cofactor

12 m_p (proton mass) M 11 104 5.98e-105 103 103-digit cofactor

13 m_p/m_e (mass ratio) M 13 105 2.41e-105 104 104-digit cofactor

14 R_inf (Rydberg constant) M 13 108 5.67e-109 106 106-digit cofactor

15 a_0 (Bohr radius) M 12 91 7.46e-92 89 89-digit cofactor

16 a_e (electron g-2 anomaly) M 12 98 4.26e-99 93 93-digit cofactor

17 a_mu (muon g-2 anomaly) M 9 98 5.62e-99 97 97-digit cofactor

18 sin2_theta_W (weak mixing) M 5 101 4.39e-102 100 100-digit cofactor

19 alpha_s (strong coupling at M_Z) M 4 100 5.13e-101 99 99-digit cofactor

20 mu_0 (vacuum permeability) M 12 95 1.22e-96 94 94-digit cofactor

21 M_Z (Z boson mass) M 6 106 3.13e-107 104 104-digit cofactor

22 Gamma_Z (Z total width) M 5 105 2.29e-105 99 99-digit cofactor

23 M_W (W boson mass) M 6 106 7.11e-107 105 105-digit cofactor

24 m_t (top quark mass) M 5 107 0.00e+00 5 5-digit cofactor

25 m_H (Higgs boson mass) M 5 106 0.00e+00 3 3-digit cofactor

26 sigma0_had (peak hadronic xsec) M 5 103 2.76e-105 103 103-digit cofactor

27 R_l (Gamma_had/Gamma_l) M 5 103 1.76e-103 101 101-digit cofactor

28 R_b (Gamma_bb/Gamma_had) M 5 101 1.67e-102 98 98-digit cofactor

29 A_FB_l (fwd-bwd asym lepton) M 3 100 4.12e-100 98 98-digit cofactor

30 A_l_SLD (polarization asym) M 4 101 2.63e-101 97 97-digit cofactor

31 N_nu (neutrino count from Z) M 5 102 5.36e-103 101 101-digit cofactor

32 G_F (Fermi constant) M 8 96 1.60e-97 95 95-digit cofactor

33 m_u (up quark MS-bar 2GeV) M 3 102 7.94e-103 99 99-digit cofactor

34 m_d (down quark MS-bar 2GeV) M 3 102 1.22e-102 101 101-digit cofactor

35 m_s (strange MS-bar 2GeV) M 3 103 0.00e+00 0 FULLY FACTORED

36 m_c (charm MS-bar at m_c) M 4 104 0.00e+00 0 FULLY FACTORED

37 m_b (bottom MS-bar at m_b) M 4 105 0.00e+00 0 FULLY FACTORED

38 sin_theta12 (Cabibbo) M 5 101 1.50e-102 98 98-digit cofactor

39 sin_theta23 (CKM) M 4 100 1.47e-100 98 98-digit cofactor

40 sin_theta13 (CKM) M 4 99 2.87e-100 95 95-digit cofactor

41 m_c/m_s (lattice ratio) M 5 102 1.75e-103 101 101-digit cofactor

42 m_b/m_c (lattice ratio) M 4 102 9.24e-103 102 102-digit cofactor

43 m_u/m_d (lattice ratio) M 3 101 1.41e-101 99 99-digit cofactor

44 m_n (neutron mass) M 11 104 7.54e-105 104 104-digit cofactor

45 m_n - m_p (mass difference) M 8 101 1.23e-102 99 99-digit cofactor

46 m_pi+ (charged pion) M 8 103 4.95e-104 103 103-digit cofactor

47 m_pi0 (neutral pion) M 7 103 4.91e-104 101 101-digit cofactor

48 m_K+ (charged kaon) M 6 104 3.94e-105 102 102-digit cofactor

49 m_D (deuteron mass) M 12 105 1.11e-105 105 105-digit cofactor

50 m_He4 (helium-4 mass) M 10 105 1.75e-105 103 103-digit cofactor

51 E_D (deuteron binding energy) M 8 102 2.74e-102 99 99-digit cofactor

52 H 1S-2S transition M 16 117 0.00e+00 14 14-digit cofactor

53 H hyperfine 1S M 13 110 3.78e-111 109 109-digit cofactor

54 Sr-87 clock transition M 16 116 0.00e+00 14 14-digit cofactor

55 Yb-171 clock transition M 16 116 5.51e-117 114 114-digit cofactor

56 Lamb shift 2S-2P1/2 M 8 107 0.00e+00 5 5-digit cofactor

57 r_p (proton charge radius) M 5 101 1.77e-102 101 101-digit cofactor

58 pi A 105 102 9.14e-103 102 102-digit cofactor

59 e (Euler's number) A 105 102 1.17e-102 100 100-digit cofactor

60 ln(2) A 105 101 6.97e-102 99 99-digit cofactor

61 R2 = pi/4 A 105 101 8.18e-102 101 101-digit cofactor

62 R4 = pi^2/32 A 105 101 1.12e-101 100 100-digit cofactor

63 2$pi = 8$R2 A 105 102 9.14e-103 102 102-digit cofactor

64 zeta(3) (Apery) A 105 101 2.51e-102 98 98-digit cofactor

65 zeta(5) A 105 101 5.67e-102 101 101-digit cofactor

66 sqrt(2) (Koide amplitude) A 105 101 6.44e-104 100 100-digit cofactor

67 sqrt(3) A 105 102 3.95e-102 101 101-digit cofactor

68 phi (golden ratio) A 105 102 4.34e-102 101 101-digit cofactor

69 gamma (Euler-Mascheroni) A 105 101 7.58e-102 100 100-digit cofactor

70 C_c = pi/(pi+2) (vena contracta) A 105 101 1.32e-102 99 99-digit cofactor

71 BCS gap = pi/exp(gamma) A 105 102 3.65e-102 101 101-digit cofactor

72 j11 (first zero of J1) A 105 102 1.31e-102 102 102-digit cofactor

73 j11/pi (Airy constant 1.22) A 105 101 2.71e-102 101 101-digit cofactor

74 j01 (first zero of J0) = 2.405 A 105 102 1.64e-102 100 100-digit cofactor

75 alpha/pi (QED expansion param) M 12 99 1.04e-99 99 99-digit cofactor

76 g_n (standard gravity) E 6 102 2.14e-103 99 99-digit cofactor

77 WGS84 semi-major axis E 7 108 0.00e+00 7 7-digit cofactor

78 WGS84 inverse flattening 1/f E 12 104 8.42e-107 100 100-digit cofactor

79 P_0 (standard atmosphere) E 6 106 0.00e+00 3 3-digit cofactor

80 A4 (concert pitch) E 3 104 0.00e+00 0 FULLY FACTORED

81 CD sample rate E 5 106 0.00e+00 0 FULLY FACTORED

82 inch (exact definition) E 3 103 1.12e-103 103 103-digit cofactor

83 p_0 (SPL reference) E 1 97 3.20e-97 96 96-digit cofactor

84 Cu conductivity 100% IACS E 5 109 0.00e+00 4 4-digit cofactor

85 CD laser wavelength S 3 95 7.76e-96 95 95-digit cofactor

86 DVD laser wavelength S 3 95 1.06e-95 91 91-digit cofactor

87 Blu-ray laser wavelength S 3 95 7.85e-96 93 93-digit cofactor

88 CD track pitch S 2 96 3.18e-96 91 91-digit cofactor

89 DVD track pitch S 2 95 1.65e-96 95 95-digit cofactor

90 Blu-ray track pitch S 3 95 3.18e-96 91 91-digit cofactor

91 Blu-ray NA S 2 101 3.36e-102 101 101-digit cofactor

92 DVD NA S 2 101 4.76e-102 101 101-digit cofactor

93 CD pit depth S 3 94 2.46e-95 91 91-digit cofactor

94 Blu-ray min pit 2T S 3 95 7.61e-96 90 90-digit cofactor

95 SMF-28 MFD at 1310nm S 2 96 7.11e-98 93 93-digit cofactor

96 SMF-28 NA S 2 100 4.90e-101 98 98-digit cofactor

97 SMF-28 cladding diameter S 4 97 3.38e-98 97 97-digit cofactor

98 Si lattice constant M 10 92 1.35e-93 89 89-digit cofactor

99 m_mu/m_e M 10 104 5.82e-105 102 102-digit cofactor

100 m_tau/m_e M 6 105 8.94e-106 104 104-digit cofactor

101 m_tau/m_mu M 6 103 2.00e-103 102 102-digit cofactor

102 m_p/m_e (direct measurement) M 13 105 2.41e-105 104 104-digit cofactor

103 m_n/m_p M 11 101 5.70e-102 100 100-digit cofactor

104 M_W/M_Z M 6 101 7.34e-102 97 97-digit cofactor

105 m_H/M_Z M 5 101 4.62e-102 100 100-digit cofactor

106 m_t/M_Z M 5 102 3.29e-103 98 98-digit cofactor

107 Koide ratio K(e,mu,tau) M 6 101 3.10e-102 98 98-digit cofactor

Total entries: 107

Type A (Analytical): 17

Type E (Exact defined): 16

Type M (Measured): 61

Type S (Standard nominal): 13

Fully factored into primes <= 97: 5/107

Precision tiers:

10+ source digits (high value for pattern search): 41

5-9 source digits (moderate value): 37

<5 source digits (low value, keep for completeness): 29

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FULLY FACTORED Q335 NUMERATORS (primes <= 97 only)

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35 m_s (strange MS-bar 2GeV) = 2^334 x 11 x 17

36 m_c (charm MS-bar at m_c) = 2^335 x 19 x 67

37 m_b (bottom MS-bar at m_b) = 2^335 x 47 x 89

80 A4 (concert pitch) = 2^338 x 5 x 11

81 CD sample rate = 2^337 x 3^2 x 5^2 x 7^2

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ENTRIES WITH SMALL COFACTORS (< 30 digits, not fully factored)

These are candidates for further factorization analysis

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1 c (speed of light) cofactor: 6 digits cofactor = 293339 x 2^336 x 7 x 73

5 N_A (Avogadro) cofactor: 9 digits cofactor = 150553519 x 2^352 x 5^15

6 dv_Cs (Cs-133 hyperfine) cofactor: 5 digits cofactor = 44351 x 2^336 x 3^2 x 5 x 7^2 x 47

7 K_cd (luminous efficacy) cofactor: 3 digits cofactor = 683 x 2^335

24 m_t (top quark mass) cofactor: 5 digits cofactor = 17257 x 2^336 x 5

25 m_H (Higgs boson mass) cofactor: 3 digits cofactor = 313 x 2^339 x 5^2

52 H 1S-2S transition cofactor: 14 digits cofactor = 33325154232257 x 2^336 x 37

54 Sr-87 clock transition cofactor: 14 digits cofactor = 25248706131169 x 2^335 x 17

56 Lamb shift 2S-2P1/2 cofactor: 5 digits cofactor = 70523 x 2^335 x 3 x 5

77 WGS84 semi-major axis cofactor: 7 digits cofactor = 6378137 x 2^335

79 P_0 (standard atmosphere) cofactor: 3 digits cofactor = 193 x 2^335 x 3 x 5^2 x 7

84 Cu conductivity 100% IACS cofactor: 4 digits cofactor = 1871 x 2^338 x 5^3 x 31

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ENTRIES WITH ZERO RECONSTRUCTION ERROR (exact in Q335)

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1 c (speed of light) factors: 2^336 x 7 x 73 6-digit cofactor

5 N_A (Avogadro) factors: 2^352 x 5^15 9-digit cofactor

6 dv_Cs (Cs-133 hyperfine) factors: 2^336 x 3^2 x 5 x 7^2 x 47 5-digit cofactor

7 K_cd (luminous efficacy) factors: 2^335 3-digit cofactor

24 m_t (top quark mass) factors: 2^336 x 5 5-digit cofactor

25 m_H (Higgs boson mass) factors: 2^339 x 5^2 3-digit cofactor

35 m_s (strange MS-bar 2GeV) factors: 2^334 x 11 x 17 FULLY FACTORED

36 m_c (charm MS-bar at m_c) factors: 2^335 x 19 x 67 FULLY FACTORED

37 m_b (bottom MS-bar at m_b) factors: 2^335 x 47 x 89 FULLY FACTORED

52 H 1S-2S transition factors: 2^336 x 37 14-digit cofactor

54 Sr-87 clock transition factors: 2^335 x 17 14-digit cofactor

56 Lamb shift 2S-2P1/2 factors: 2^335 x 3 x 5 5-digit cofactor

77 WGS84 semi-major axis factors: 2^335 7-digit cofactor

79 P_0 (standard atmosphere) factors: 2^335 x 3 x 5^2 x 7 3-digit cofactor

80 A4 (concert pitch) factors: 2^338 x 5 x 11 FULLY FACTORED

81 CD sample rate factors: 2^337 x 3^2 x 5^2 x 7^2 FULLY FACTORED

84 Cu conductivity 100% IACS factors: 2^338 x 5^3 x 31 4-digit cofactor

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Q335 CONVERSION COMPLETE

2^335 has 101 decimal digits

This basis can represent ~100 decimal digits of precision.

All entries with source_digits <= 100 are faithfully represented.

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[**HOWL-DATA-2-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/DATA/I
DATA-2-2026

[**HOWL-DATA-1-2026**] Paternal Operationalism. Github: https://github.com/ghowland/papers/tree/main/papers/DATA/I
DATA-1-2026